CMB constraints on compact baryonic clumps as part of the dark matter: summary for discussion

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1 Purpose

This note summarises what the Planck CMB data say about compact, pressureless clumps of baryonic matter that make up some or all of the dark matter, and where clouds of the kind proposed by Walker & Wardle (1998) fall in that picture. The background physics is set out in the explainer (`doc/explainer/clumps_explainer.pdf`) and the analysis for clumps that make up all of the dark matter in the report (`doc/report/cmb_bound_report.pdf`). This note adds the case in which only a fraction of the dark matter is in clumps.

2 The question and the model

The question is conditional: if clumps exist throughout the epochs that matter for the CMB, how strongly can they be coupled to the radiation? Formation, survival and Big Bang nucleosynthesis (BBN) are separate questions about whether such clumps can exist, and are discussed in Sections 5–7.

The clumps exchange momentum with the photons through a momentum-transfer cross-section σ. The CMB is sensitive only to the cross-section per unit mass, expressed in CLASS by the dimensionless coupling

$$u = \frac{\sigma}{\sigma_T}\left(\frac{M}{100\,\mathrm{GeV}}\right)^{-1}, \qquad \frac{\sigma}{M} = \frac{u}{268}\,\mathrm{cm}^2\,\mathrm{g}^{-1}. \tag{1}$$

For an optically thick clump $\sigma = Q\,\pi R^2$ with Q of order unity ($Q = 1$ for an absorber), so a limit on u is a limit on the surface density,

$$\frac{\Sigma}{Q} = \frac{M}{Q\,\pi R^2} = \frac{268}{u}\,\mathrm{g\,cm}^{-2}. \tag{2}$$

The CMB signal comes from radiation drag on the clumps before recombination, which suppresses the growth of their density perturbations. Planck sees this mainly through reduced gravitational lensing of the CMB. Photons scattering off the clumps is a negligible effect at $f_{\rm cl} = 1$ but becomes important at small $f_{\rm cl}$ and strong coupling (Section 4).

For the mixture, a fraction $f_{\rm cl}$ of the dark matter is in clumps and the rest is collisionless cold dark matter (CDM):

$$\omega_{\rm cl} = f_{\rm cl}\,\omega_{\rm dark}, \qquad \omega_{\rm CDM} = (1 - f_{\rm cl})\,\omega_{\rm dark}. \tag{3}$$

The total $\omega_{\rm dark}$, the diffuse baryon density ω_b, the other four ΛCDM parameters and the nine likelihood nuisance parameters are all fitted. Two angular laws for the photon–clump interaction are compared at equal u: Thomson scattering, which generates polarization, and uniform re-emission in the clump frame with no polarization, which is the appropriate limit for an opaque body.

Table 1: Nominal fixed-fraction thresholds ($\Delta\chi^2 = 2.71$ above each fraction's own profile minimum), PR4 CamSpec with 2018 low-ℓ. "Not reached" means no threshold within the computed range, whose smallest Σ/Q is given in brackets. The last column is the depth of the profile minimum relative to ΛCDM for the isotropic kernel.

$f_{\rm cl}$	Σ/Q [g cm^{-2}], Thomson	Σ/Q [g cm^{-2}], isotropic	u (Thomson)	min. − ΛCDM
1	2.4×10^6	2.3×10^6	1.1×10^{-4}	0
0.5	1.0×10^6	9.8×10^5	2.7×10^{-4}	0
0.1	6.6×10^4	6.7×10^4	4.1×10^{-3}	0
0.05	1.4×10^4	1.5×10^4	1.9×10^{-2}	−0.3
0.03	7.1×10^3	7.1×10^3	3.8×10^{-2}	−1.8
0.02	4.3×10^3	4.0×10^3	6.2×10^{-2}	−2.6
0.01	1.8×10^3	9.6×10^2	0.15	−2.9
0.005	not reached ($> 1.0 \times 10^3$)	not reached ($> 4.5 \times 10^2$)	–	−2.8
0.003	not reached ($> 6.0 \times 10^2$)	not reached ($> 2.7 \times 10^2$)	–	−2.5
0.001	not reached ($> 2.0 \times 10^2$)	not reached ($> 0.9 \times 10^2$)	–	−1.2

3 Data and method

The likelihood is the Planck PR4 CamSpec high-ℓ TT, TE and EE likelihood with the Planck 2018 low-ℓ TT and EE likelihoods. The spectra are computed with CLASS 3.4.0, patched to allow the second kernel, and with tighter numerical settings than the defaults. For each fixed $(f_{\rm cl}, u)$ the objective is minimised over the other fifteen parameters (a profile likelihood). Results are given as

- $\Delta\chi^2$ relative to the best ΛCDM fit, and
- nominal fixed-fraction thresholds: the coupling at which $\Delta\chi^2$ rises 2.71 above that fraction's own minimum. These are not joint confidence regions in $(f_{\rm cl}, u)$.

About 150 profile points were computed. Repeated fits agree to about 0.02 in $\Delta\chi^2$. Changing the tight-coupling approximation or the numerical precision changes χ^2 by at most about 0.1 relative to ΛCDM at every point checked, including the strongest couplings.

4 Results

4.1 All of the dark matter in clumps

From the report: $u < 1.1 \times 10^{-4}$ (profile, PR4 CamSpec; 1.0×10^{-4} with Planck 2018 plik), and $u < 1.7 \times 10^{-4}$ from a Bayesian analysis with a flat prior on u. These correspond to $\Sigma/Q \gtrsim 2 \times 10^6\,\mathrm{g\,cm^{-2}}$, about the column density of ten kilometres of rock. The kernel changes this by about 3%.

4.2 A fraction of the dark matter in clumps

Table 1 and Figure 1 summarise the results. Three points stand out.

CDM weakens the constraint, faster than $1/f_{\rm cl}$. The drag suppresses only the clump perturbations, and after decoupling the CDM pulls the clumps back into its potential wells. The threshold on u rises by a factor of 36 from $f_{\rm cl} = 1$ to 0.1, and by about 1300 to $f_{\rm cl} = 0.01$. The minimum surface density falls correspondingly from 2×10^6 to about $10^3\,\mathrm{g\,cm^{-2}}$.

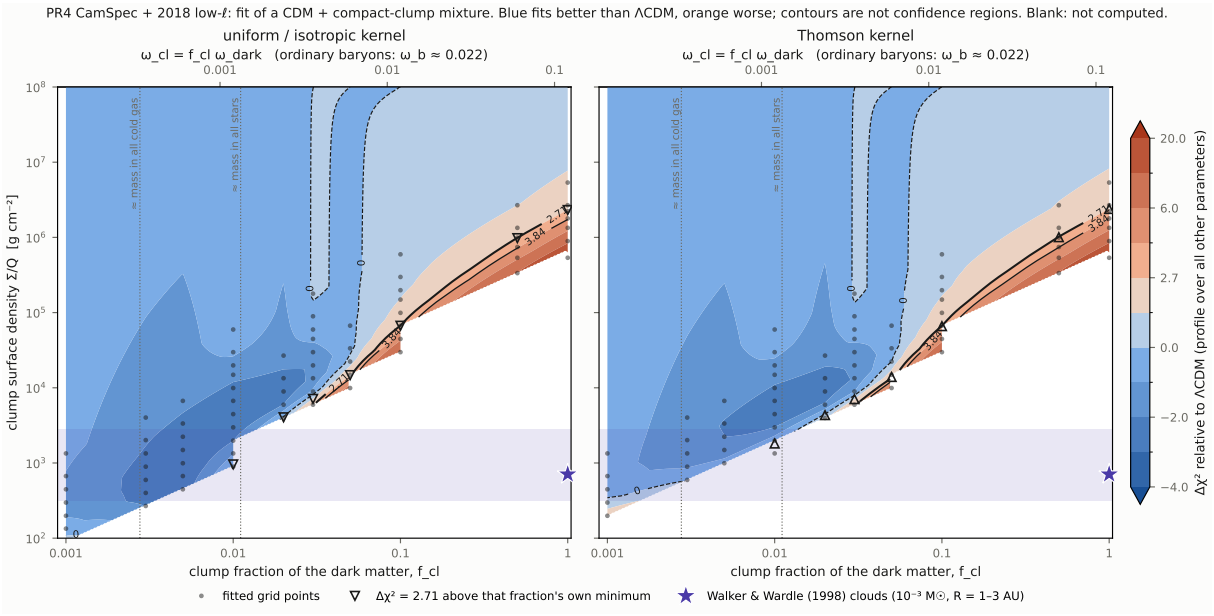


Figure 1: $\Delta\chi^2$ relative to ΛCDM as a function of clump fraction and surface density, for the isotropic (left) and Thomson (right) kernels. Blue regions fit as well as or slightly better than ΛCDM, orange regions worse. Triangles mark the fixed-fraction thresholds of Table 1. The shaded band and star show the surface density of the clouds of Walker & Wardle (1998). Dotted lines mark clump densities equal to the present-day mass in stars and in cold gas, as approximate fractions of the cosmic baryons. Blank regions were not computed.

The kernel matters only at small fractions and strong coupling. For $f_{\rm cl} \geq 0.02$ the two kernels give thresholds within about 7%. For $f_{\rm cl} \lesssim 0.01$ the required coupling is large enough that the clumps carry an appreciable part of the photon scattering after recombination, and Thomson scattering off them generates polarization that the data disfavour. The isotropic kernel, which is the appropriate one for opaque clumps, allows lower surface densities there.

A shallow minimum below ΛCDM. For $f_{\rm cl} \approx 0.003$–0.02 the profiles have a minimum at non-zero coupling, about 2.5 to 2.9 below ΛCDM in χ^2. The improvement comes from the low-ℓ temperature spectrum and is accompanied by a higher spectral index ($n_s \approx 0.97$ instead of 0.961). It is the known deficit of Planck's low-ℓ temperature power, which other one-parameter extensions also fit. With two fitted parameters the significance is about 1.2σ ($p \approx 0.23$), lower once the search over the valley is accounted for. It is not evidence for clumps.

4.3 Where the Walker & Wardle clouds fall

Clouds with $M \lesssim 10^{-3}\,M_\odot$ and radii of a few AU (Walker & Wardle, 1998) have $\Sigma \approx 3 \times 10^2$–$3 \times 10^3\,{\rm g\,cm^{-2}}$. On the CMB evidence alone:

- as all of the dark matter they are excluded by more than three orders of magnitude in Σ;
- with the isotropic kernel they are allowed for $f_{\rm cl} \lesssim 0.01$, and lie in the region that fits as well as ΛCDM;
- $f_{\rm cl} \approx 0.003$–0.01 corresponds to $\omega_{\rm cl} \approx (0.4$–$1.2) \times 10^{-3}$, or 1.5–5% of the cosmic baryons, comparable to the present-day mass in cold gas or in stars. In a Milky-Way halo of $10^{12}\,M_\odot$ it would be $(0.3$–$1) \times 10^{10}\,M_\odot$ in clouds.

5 Thermal state of the clumps

Before recombination the radiation field is a blackbody at T_γ, about 3000 K at recombination, 10^4 K at $z \approx 4000$ and 10^5 K at $z \approx 4 \times 10^4$. A passive clump can lose heat only by radiating into this field, so its steady-state temperature is T_γ. Emissivity, albedo and composition change the time taken to reach it, not the end state. A clump can be colder than the radiation only transiently, while its interior lags, or if it interacts with the radiation so weakly that it is barely heated. The relevant timescales are as follows.

- *Surface.* The surface relaxes to T_γ on $t \approx \Sigma c_v/(4caT_\gamma^3)$, short enough that it lags the cooling CMB by about 10^{-8} of T_γ at the surface densities of interest (explainer, Section 9). Heat capacity and latent heat (dissociation of H_2, ionisation of H) lengthen this by at most about ten. Only absorbed photons deposit energy, so a reflective surface heats more slowly by $1/(1-A)$ for albedo A, which matters only for A above about 0.999999.

- *Interior.* Radiative diffusion across a cloud takes $t \approx \kappa\Sigma R/c$. For $\Sigma = 10^3\,\mathrm{g\,cm^{-2}}$ and $R = 2\,\mathrm{AU}$ this is days with electron-scattering opacity and decades with an opacity 10^4 times larger, compared with 10^4–10^5 years for the age of the universe at these epochs. A cold interior is not maintained.

- *Transparent clouds.* A cloud transparent to the ambient radiation, such as dust-free molecular hydrogen in parts of the spectrum, is heated only weakly and could remain cold. The same transparency removes its coupling to the radiation, so it is not the opaque object assumed in the Σ/Q mapping, and the drag constraint does not apply to it. The photons that would push a clump are the same photons that heat it.

Whether a heated clump survives depends on what holds it together. Solids sublimate at roughly 1500–3000 K, so no solid body survives before $z \approx 1000$. Gas at T_γ is bound only if $GM\mu m_p/R \gtrsim kT_\gamma$, so composition enters through the mean particle mass μ, and only linearly. For a cloud of $10^{-3}\,M_\odot$ and $R = 2\,\mathrm{AU}$ the binding temperature is about 100 K for molecular hydrogen, which allows binding only below $z \approx 50$. A cloud made entirely of iron vapour would gain a factor of about 25, still only reaching $z \approx 1000$. Degenerate or nuclear-density objects are bound independently of temperature and survive. Internal energy sources, such as contraction or nuclear burning, heat the interior and do not help a cold-cloud picture.

Today the background radiation is at 2.7 K, and clouds such as those of Walker & Wardle (1998) can be colder than the surrounding halo gas. That is the regime in which the model was proposed. Before $z \approx 50$ the radiation itself is hotter than the clouds' binding temperature.

6 Formation: recent work on small-scale baryon structure

Primordial magnetic fields generate baryon density contrasts of order unity on small scales before recombination, which accelerate recombination because the recombination rate depends on the square of the density (Jedamzik & Saveliev, 2019). Simulations combining magnetohydrodynamics, recombination and Lyman-α transfer (Jedamzik et al., 2023), and fits of the resulting recombination histories to Planck, baryon acoustic oscillation and supernova data (Jedamzik et al., 2025), favour present-day field strengths of roughly 5–10 pG. This work bears on the clump question in three ways.

- *The inhomogeneities are not compact.* They are diffuse fluctuations on scales of order a comoving kiloparsec, a few parsecs physical at $z \sim 1000$, containing of order a solar mass of baryons each by our estimate. Their column densities are of order $10^{-3}\,\mathrm{g\,cm^{-2}}$, nine orders of magnitude below the surface densities discussed above. They remain part of the photon–baryon fluid and are counted in ω_b, so they provide no way to hide baryons from the CMB.

- *They provide seeds for later formation.* Overdense regions recombine and decouple from the photons slightly earlier, and at the favoured field strengths magnetic pressure is comparable to gas pressure on small scales. Clouds formed from such seeds after $z \approx 50$ would be drawn from the measured baryon budget: a component of the ordinary baryons, not of the dark matter, and not subject to the drag constraint. Whether field-seeded overdensities can collapse into objects of $10^{-3}\,M_\odot$ and a few AU before $z \sim 20$–50, and what fraction of the baryons could end up in them within the existing field limits ($B \lesssim 50\,\mathrm{pG}$; Jedamzik & Saveliev 2019), is a calculation we have not yet made.

- *Modified recombination is a separate change to the CMB model.* It shifts the sound horizon and the damping tail, as the clumps at small $f_{\rm cl}$ do. A fit with both could move the thresholds and the shallow minimum in Section 4 in either direction. The published recombination histories could be included in our pipeline.

Older work on nucleosynthesis with large baryon inhomogeneities, which would be the relevant framework for clumps formed before BBN, concluded as far as we recall that such inhomogeneities do not substantially relax the bound on the total baryon density. This needs to be checked.

7 Caveats

Existence through the acoustic epoch. All of the above assumes the clumps exist, with the stated Σ, at redshifts of roughly 10^3–10^5. For clouds like those of Walker & Wardle (1998) this assumption fails: such a cloud can be gravitationally bound only below $z \approx 50$ (Section 5). If such clouds form later, from gas that was diffuse at recombination, the CMB counts that gas as ordinary baryons and the drag constraint does not apply to them at all. The relevant constraint is then the baryon budget: the clouds must come out of the measured ω_b, and so can be at most a fraction of the ordinary baryons, not a separate dark component.

Nucleosynthesis. Clumps formed after BBN are counted as baryons by the light-element abundances, while the CMB counts them as dark matter. The agreement of the BBN and CMB baryon densities leaves room for an extra component of only a few per cent of ω_b, which corresponds to $f_{\rm cl} \lesssim 0.01$. This figure depends on the nuclear reaction rates adopted and should be checked.

Scope of the model. The limits hold only if the clumps are the sole departure from ΛCDM. A population of early compact objects would probably come with other changes, such as small-scale clumping of the remaining gas and a modified recombination history, that are not modelled and could move the limits in either direction.

Statistics. The thresholds are fixed-fraction profile values. A Bayesian analysis with $f_{\rm cl}$ and u both free is planned. At $f_{\rm cl} = 1$ it gave a limit 50% weaker than the profile, so the boundary in Figure 1 may move by a similar factor.

8 Points for discussion

1. At what redshift do the clouds need to exist in the proposed scenario, and from what material do they form? Could small-scale structure seeded by primordial magnetic fields (Section 6) form them after $z \approx 50$?

2. Is the intended cloud population within the baryon budget fixed by BBN and the CMB, or in addition to it?

3. Does the scattering-event model fix the cloud surface density independently? If so, the CMB constraint on $f_{\rm cl}$ at that fixed Σ becomes a one-parameter test.

4. Which angular law is appropriate for photon interactions with the clouds at the relevant epochs: absorption and isotropic re-emission, or something closer to reflection?

References

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